

SQL STUDENT DATABASE ANALYSIS PROJECT

PART-2

QUERIES WITH OUTPUT
(31-60)

TOOL USED: MYSQL
BY SHAGUN



OBJECTIVE

- This project aims to strengthen SQL skills by solving real-world database queries using an Employee Management Database. The project consists of 100 SQL questions covering both basic and advanced concepts.

Current Progress

- Total Questions: 100
- Completed questions: 60
- Questions Solved: 30

BASIC SQL CONCEPTS IMPLEMENTED

- GROUP BY
- HAVING Clause
- WHERE Clause
- DISTINCT
- ORDER BY
- LIMIT
- Aggregate Functions (COUNT, SUM, AVG, MIN, MAX)

CREATE & USE DATABASE

```
1  -- Count students in each course
2  • SELECT Course, COUNT(*) AS TotalStudents
3     FROM Student
4     GROUP BY Course;
```

Result Grid			Filter Rows:	Export:	Write
	Course	TotalStudents			
▶	BCA	10			
	BBA	8			
	B.Tech	7			
	MBA	9			
	MCA	6			
	B.Sc	5			
	B.Com	5			

CREATE TABLE

```
6      -- Count students by gender
7  •   SELECT GENDER, count(*) AS ToTalStudents
8      FROM student
9      GROUP BY Gender;
```

Result Grid



Filter Rows:

Export:



W

	GENDER	ToTalStudents
▶	Male	25
	Female	25

INSERT DATA

```
13      -- Find average age for each course
14 •    SELECT Course, AVG(Age) AS AverageAge
15      FROM Student
16      GROUP BY Course;
17
```

Result Grid			Filter Rows:	Export:
	Course	AverageAge		
▶	BCA	20.1000		
	BBA	21.1250		
	B.Tech	20.4286		
	MBA	21.8889		
	MCA	21.5000		
	B.Sc	20.6000		
	B.Com	21.0000		

DISPLAY ALL RECORDS

```
17      -- Find oldest student age in each course
18  •  SELECT Course, MAX(Age) AS OldestAge
19      FROM Student
20      GROUP BY Course;
21
```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Contents: ☐

	Course	OldestAge
▶	BCA	21
	BBA	22
	B.Tech	22
	MBA	23
	MCA	23
	B.Sc	22
	B.Com	23

DISPLAY ONLY FIRSTNAME & LASTNAME

```
22      -- Find youngest student age in each course
23      SELECT Course, min(age) AS YoungestAge
24      FROM student
25      GROUP BY Course;
26
```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Cont

	Course	YoungestAge
▶	BCA	19
	BBA	20
	B.Tech	19
	MBA	20
	MCA	20
	B.Sc	19
	B.Com	19

SHOW ALL UNIQUE COURSES

```
27      -- Display courses with student counts in descending order
28  •  SELECT Course, COUNT(*) AS TotalStudents
29      FROM Student
30      GROUP BY Course
31      ORDER BY TotalStudents DESC;
```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	Course	TotalStudents
▶	BCA	10
	MBA	9
	BBA	8
	B.Tech	7
	MCA	6
	B.Sc	5
	B.Com	5

FIND ALL MALE STUDENTS

```
33 -- Find students whose first name contains 'a'
34 • SELECT * FROM student
35 where FirstName LIKE '%a%';
```

Result Grid

 Filter Rows:

Edit:   

Export/Import:  

Wrap Cell Content: 

	StudentID	FirstName	LastName	Gender	Age	Email	PhoneNumber	Course	AdmissionDate
▶	1	Rahul	Sharma	Male	20	rahul@email.com	9876543210	BCA	2025-06-01
	2	Priya	Verma	Female	21	priya@email.com	9876543211	BBA	2025-06-02
	3	Amit	Kumar	Male	19	amit@email.com	9876543212	B.Tech	2025-06-03
	4	Neha	Singh	Female	22	neha@email.com	9876543213	MBA	2025-06-04
	6	Anjali	Patel	Female	21	anjali@email.com	9876543215	B.Sc	2025-06-06
	7	Vikas	Mishra	Male	23	vikas@email.com	9876543216	B.Com	2025-06-07
	8	Sneha	Chauhan	Female	20	sneha@email.com	9876543217	BCA	2025-06-08
	9	Arjun	Mehta	Male	22	arjun@email.com	9876543218	B.Tech	2025-06-09
	10	Kavita	Agarwal	Female	21	kavita@email.com	9876543219	MBA	2025-06-10
	11	Saurabh	Jain	Male	20	saurabh@email.com	9876543220	BBA	2025-06-11
	12	Pooja	Yadav	Female	19	pooja@email.com	9876543221	BCA	2025-06-12
	13	Deepak	Tiwari	Male	22	deepak@email.com	9876543222	MCA	2025-06-13
	14	Riya	Saxena	Female	20	riya@email.com	9876543223	B.Com	2025-06-14
	15	Manish	Pandey	Male	23	manish@email.com	9876543224	MBA	2025-06-15
	16	Nisha	Joshi	Female	21	nisha@email.com	9876543225	B.Sc	2025-06-16
	17	Karan	Malhotra	Male	20	karan@email.com	9876543226	B.Tech	2025-06-17

FIND ALL FEMALE STUDENTS

```
38      -- Find students whose last name contains 'h'
39  •  SELECT *
40      FROM Student
41      WHERE LastName LIKE '%h%';
```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:									
	StudentID	FirstName	LastName	Gender	Age	Email	PhoneNumber	Course	AdmissionDate
▶	1	Rahul	Sharma	Male	20	rahul@email.com	9876543210	BCA	2025-06-01
	4	Neha	Singh	Female	22	neha@email.com	9876543213	MBA	2025-06-04
	7	Vikas	Mishra	Male	23	vikas@email.com	9876543216	B.Com	2025-06-07
	8	Sneha	Chauhan	Female	20	sneha@email.com	9876543217	BCA	2025-06-08
	9	Arjun	Mehta	Male	22	arjun@email.com	9876543218	B.Tech	2025-06-09
	16	Nisha	Joshi	Female	21	nisha@email.com	9876543225	B.Sc	2025-06-16
	17	Karan	Malhotra	Male	20	karan@email.com	9876543226	B.Tech	2025-06-17
	21	Harsh	Sinha	Male	20	harsh@email.com	9876543230	BCA	2025-06-21
	22	Ayesha	Khan	Female	22	ayesha@email.com	9876543231	BBA	2025-06-22
	23	Nitin	Rathore	Male	21	nitin@email.com	9876543232	B.Tech	2025-06-23
	25	Varun	Shukla	Male	23	varun@email.com	9876543234	MBA	2025-06-25
	27	Akash	Tripathi	Male	22	akash@email.com	9876543236	B.Sc	2025-06-27
	28	Tarvi	Mahajan	Female	20	tarvi@email.com	9876543237	BCA	2025-06-28
	29	Mohit	Bhardwaj	Male	21	mohit@email.com	9876543238	BBA	2025-06-29
	30	Komal	Sharma	Female	22	komal@email.com	9876543239	MBA	2025-06-30
	34	Ishita	Mishra	Female	20	ishita@email.com	9876543243	BBA	2025-07-04

FIND STUDENTS OLDER THAN 20

```
43 -- Find students aged 20 or more and enrolled in BCA
44 • SELECT * FROM Student
45 WHERE Age >= 20
46 AND Course = 'BCA';
```

	StudentID	FirstName	LastName	Gender	Age	Email	PhoneNumber	Course	AdmissionDate
►	1	Rahul	Sharma	Male	20	rahul@email.com	9876543210	BCA	2025-06-01
	4	Neha	Singh	Female	22	neha@email.com	9876543213	MBA	2025-06-04
	7	Vikas	Mishra	Male	23	vikas@email.com	9876543216	B.Com	2025-06-07
	8	Sneha	Chauhan	Female	20	sneha@email.com	9876543217	BCA	2025-06-08
	9	Arjun	Mehta	Male	22	arjun@email.com	9876543218	B.Tech	2025-06-09
	16	Nisha	Joshi	Female	21	nisha@email.com	9876543225	B.Sc	2025-06-16
	17	Karan	Malhotra	Male	20	karan@email.com	9876543226	B.Tech	2025-06-17
	21	Harsh	Sinha	Male	20	harsh@email.com	9876543230	BCA	2025-06-21
	22	Ayesha	Khan	Female	22	ayesha@email.com	9876543231	BBA	2025-06-22
	23	Nitin	Rathore	Male	21	nitin@email.com	9876543232	B.Tech	2025-06-23
	25	Varun	Shukla	Male	23	varun@email.com	9876543234	MBA	2025-06-25
	27	Akash	Tripathi	Male	22	akash@email.com	9876543236	B.Sc	2025-06-27
	28	Tanvi	Mahajan	Female	20	tanvi@email.com	9876543237	BCA	2025-06-28
	29	Mohit	Bhardwaj	Male	21	mohit@email.com	9876543238	BBA	2025-06-29
	30	Komal	Sharma	Female	22	komal@email.com	9876543239	MBA	2025-06-30
	34	Ishita	Mishra	Female	20	ishita@email.com	9876543243	BBA	2025-07-04

FIND STUDENTS YOUNGER THAN 21

```
48      -- Find female students older than 20
49  •   SELECT * FROM Student
50      WHERE Gender = 'Female' AND Age > 20;
```

[illegible]

FIND STUDENTS WHOSE AGE IS EXACTLY 20

```
104 -- Find students whosw age is exactly 20
105 • SELECT * FROM student
106 Where age=20;
107
```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:									
	StudentID	FirstName	LastName	Gender	Age	Email	PhoneNumber	Course	AdmissionDate
▶	1	Rahul	Sharma	Male	20	rahul@email.com	9876543210	BCA	2025-06-01
	5	Rohit	Gupta	Male	20	rohit@email.com	9876543214	MCA	2025-06-05
	8	Sneha	Chauhan	Female	20	sneha@email.com	9876543217	BCA	2025-06-08
	11	Saurabh	Jain	Male	20	saurabh@email.com	9876543220	BBA	2025-06-11
	14	Riya	Saxena	Female	20	riya@email.com	9876543223	B.Com	2025-06-14
	17	Karan	Malhotra	Male	20	karan@email.com	9876543226	B.Tech	2025-06-17
	21	Harsh	Sinha	Male	20	harsh@email.com	9876543230	BCA	2025-06-21
	24	Megha	Arora	Female	20	megha@email.com	9876543233	MCA	2025-06-24
	28	Tanvi	Mahajan	Female	20	tanvi@email.com	9876543237	BCA	2025-06-28
	31	Yash	Verma	Male	20	yash@email.com	9876543240	B.Tech	2025-07-01
	34	Ishita	Mishra	Female	20	ishita@email.com	9876543243	BBA	2025-07-04

FIND ALL STUDENTS ENROLLED IN BCA

```
52  -- Find male students or MBA students
53  SELECT * FROM Student
54  WHERE Gender = 'Male'
55  OR Course = 'MBA';
```

Result Grid									
Filter Rows:									
Edit: Export/Import: Wrap Cell Content:									
	StudentID	FirstName	LastName	Gender	Age	Email	PhoneNumber	Course	AdmissionDate
▶	1	Rahul	Sharma	Male	20	rahul@email.com	9876543210	BCA	2025-06-01
	3	Amit	Kumar	Male	19	amit@email.com	9876543212	B.Tech	2025-06-03
	4	Neha	Singh	Female	22	neha@email.com	9876543213	MBA	2025-06-04
	5	Rohit	Gupta	Male	20	rohit@email.com	9876543214	MCA	2025-06-05
	7	Vikas	Mishra	Male	23	vikas@email.com	9876543216	B.Com	2025-06-07
	9	Arjun	Mehta	Male	22	arjun@email.com	9876543218	B.Tech	2025-06-09
	10	Kavita	Agarwal	Female	21	kavita@email.com	9876543219	MBA	2025-06-10
	11	Saurabh	Jain	Male	20	saurabh@email.com	9876543220	BBA	2025-06-11
	13	Deepak	Tiwari	Male	22	deepak@email.com	9876543222	MCA	2025-06-13
	15	Manish	Pandey	Male	23	manish@email.com	9876543224	MBA	2025-06-15
	17	Karan	Malhotra	Male	20	karan@email.com	9876543226	B.Tech	2025-06-17

SORT STUDENTS BY FIRST NAME

```
57 -- Find students not enrolled in BCA
58 • SELECT * FROM student
59 where course<>'BCA';
```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:									
	StudentID	FirstName	LastName	Gender	Age	Email	PhoneNumber	Course	AdmissionDate
▶	2	Priya	Verma	Female	21	priya@email.com	9876543211	BBA	2025-06-02
	3	Amit	Kumar	Male	19	amit@email.com	9876543212	B.Tech	2025-06-03
	4	Neha	Singh	Female	22	neha@email.com	9876543213	MBA	2025-06-04
	5	Rohit	Gupta	Male	20	rohit@email.com	9876543214	MCA	2025-06-05
	6	Anjali	Patel	Female	21	anjali@email.com	9876543215	B.Sc	2025-06-06
	7	Vikas	Mishra	Male	23	vikas@email.com	9876543216	B.Com	2025-06-07
	9	Arjun	Mehta	Male	22	arjun@email.com	9876543218	B.Tech	2025-06-09
	10	Kavita	Agarwal	Female	21	kavita@email.com	9876543219	MBA	2025-06-10
	11	Saurabh	Jain	Male	20	saurabh@email.com	9876543220	BBA	2025-06-11
	13	Deepak	Tiwari	Male	22	deepak@email.com	9876543222	MCA	2025-06-13
	14	Riya	Saxena	Female	20	riya@email.com	9876543223	B.Com	2025-06-14
	15	Manish	Pandey	Male	23	manish@email.com	9876543224	MBA	2025-06-15
	16	Nisha	Joshi	Female	21	nisha@email.com	9876543225	B.Sc	2025-06-16
	17	Karan	Malhotra	Male	20	karan@email.com	9876543226	B.Tech	2025-06-17
	18	Shreya	Kapoor	Female	22	shreya@email.com	9876543227	BBA	2025-06-18
	20	Simran	Bansal	Female	21	simran@email.com	9876543229	MBA	2025-06-20

SORT STUDENTS BY AGE IN DESCENDING ORDER

```
61  -- Count distinct courses
62  • SELECT COUNT(DISTINCT Course) AS TotalCourses
63  FROM Student;
```

Result Grid



Filter Rows:

Export:



Wrap Cell Co

	TotalCourses
▶	7

COUNT TOTAL STUDENTS

```
65  -- Display length of each student's first name
66  • SELECT FirstName, length(FirstName) AS NameLength
67  FROM student;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
FirstName	NameLength		
▶ Rahul	5		
Priya	5		
Amit	4		
Neha	4		
Rohit	5		
Anjali	6		
Vikas	5		
Sneha	5		
Arjun	5		
Kavita	6		
Saurabh	7		

FIND THE MAXIMUM AGE

```
69  -- Convert all first names to uppercase
70  • SELECT upper(FirstName) AS FirstName
71  FROM student ;
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content

	FirstName
▶	RAHUL
	PRIYA
	AMIT
	NEHA
	ROHIT
	ANJALI
	VIKAS
	SNEHA
	ARJUN
	KAVITA

FIND THE minimum AGE

```
73      -- Convert all last names to lowercase
74  ●    SELECT lower(LastName) AS LastName
75      FROM student ;
```

Result Grid



Filter Rows:

Export:



Wrap Cell Conte

	LastName
▶	sharma
	verma
	kumar
	singh
	gupta
	patel
	mishra
	chauhan
	mehta
	agarwal
	jain

FIND THE AVERAGE AGE

```
77 -- Display current date
78 SELECT CURDATE();
```

Result Grid

	CURDATE()
▶	2026-06-26

FIND THE TOTAL OF ALL AGE

```
80      -- Calculate days since admission for each student
81      SELECT FirstName,
82             DATEDIFF(CURDATE(), AdmissionDate) AS DaysSinceAdmission
83      FROM Student;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
FirstName	DaysSinceAdmission		
Rahul	390		
Priya	389		
Amit	388		
Neha	387		
Rohit	386		
Anjali	385		
Vikas	384		
Sneha	383		
Arjun	382		
Kavita	381		

DISPLAY THE FIRST 5 STUDENTS

```
85  -- Find students admitted in June
86  • SELECT * FROM Student
87  WHERE MONTH(AdmissionDate) = 6;
```

	StudentID	FirstName	LastName	Gender	Age	Email	PhoneNumber	Course	AdmissionDate
▶	1	Rahul	Sharma	Male	20	rahul@email.com	9876543210	BCA	2025-06-01
	2	Priya	Verma	Female	21	priya@email.com	9876543211	BBA	2025-06-02
	3	Amit	Kumar	Male	19	amit@email.com	9876543212	B.Tech	2025-06-03
	4	Neha	Singh	Female	22	neha@email.com	9876543213	MBA	2025-06-04
	5	Rohit	Gupta	Male	20	rohit@email.com	9876543214	MCA	2025-06-05
	6	Anjali	Patel	Female	21	anjali@email.com	9876543215	B.Sc	2025-06-06
	7	Vikas	Mishra	Male	23	vikas@email.com	9876543216	B.Com	2025-06-07
	8	Sneha	Chauhan	Female	20	sneha@email.com	9876543217	BCA	2025-06-08
	9	Arjun	Mehta	Male	22	arjun@email.com	9876543218	B.Tech	2025-06-09
	10	Kavita	Agarwal	Female	21	kavita@email.com	9876543219	MBA	2025-06-10
	11	Saurabh	Jain	Male	20	saurabh@email.c...	9876543220	BBA	2025-06-11
	12	Pooja	Yadav	Female	19	pooja@email.com	9876543221	BCA	2025-06-12

DISPLAY STUDENTS ADMITTED AFTER 15-06-2025

```
89      -- Find students admitted in 2025
90      SELECT * FROM Student
91      WHERE YEAR(AdmissionDate) = 2025;
```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content: ☐									
	StudentID	FirstName	LastName	Gender	Age	Email	PhoneNumber	Course	AdmissionDate
▶	1	Rahul	Sharma	Male	20	rahul@email.com	9876543210	BCA	2025-06-01
	2	Priya	Verma	Female	21	priya@email.com	9876543211	BBA	2025-06-02
	3	Amit	Kumar	Male	19	amit@email.com	9876543212	B.Tech	2025-06-03
	4	Neha	Singh	Female	22	neha@email.com	9876543213	MBA	2025-06-04
	5	Rohit	Gupta	Male	20	rohit@email.com	9876543214	MCA	2025-06-05
	6	Anjali	Patel	Female	21	anjali@email.com	9876543215	B.Sc	2025-06-06
	7	Vikas	Mishra	Male	23	vikas@email.com	9876543216	B.Com	2025-06-07
	8	Sneha	Chauhan	Female	20	sneha@email.com	9876543217	BCA	2025-06-08
	9	Arjun	Mehta	Male	22	arjun@email.com	9876543218	B.Tech	2025-06-09
	10	Kavita	Agarwal	Female	21	kavita@email.com	9876543219	MBA	2025-06-10

SELECT STUDENTS WHOSE FIRST NAME STARTS WITH A

```
93      -- Show courses having more than 2 students
94      SELECT Course, COUNT(*) AS TotalStudents
95      FROM Student
96      GROUP BY Course
97      HAVING COUNT(*) > 2;
```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content

	Course	TotalStudents
▶	BCA	10
	BBA	8
	B.Tech	7
	MBA	9
	MCA	6
	B.Sc	5
	B.Com	5

SELECT STUDENTS WHOSE LAST NAME ENDS WITH A

```
99      -- Find average age by gender
100      SELECT gender, avg(AGE) AS averageAGE
101      FROM student
102      GROUP BY gender;
```

Result Grid |   Filter Rows: | Export:  | Wrap

	gender	averageAGE
▶	Male	21.1600
	Female	20.7200

FIND STUDENT WITH NOT NULL

```
104      -- Show top 3 oldest students
105      SELECT * FROM Student
106      ORDER BY Age DESC
107      LIMIT 3;
```

[illegible]

FIND STUDENT WHOSE AGE IS BETWEEN 20 AND 22

```
109      -- Show top 3 youngest students
110      • select * from student
111          order by age asc
112          limit 3;
```

[illegible]

FIND ALL STUDENTS ENROLLED IN BCA OR MBA

```
120      -- Replace email domain with gmail.com
121 •    SELECT Email,
122      REPLACE(Email, '@email.com', '@gmail.com') AS NewEmail
123      FROM Student;
```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	Email	NewEmail
▶	abhishek@email.com	abhishek@gmail.com
	aditya@email.com	aditya@gmail.com
	akash@email.com	akash@gmail.com
	amit@email.com	amit@gmail.com
	anjali@email.com	anjali@gmail.com
	arjun@email.com	arjun@gmail.com
	ashish@email.com	ashish@gmail.com
	ayesha@email.com	ayesha@gmail.com
	ayush@email.com	ayush@gmail.com

FIND STUDENTS WHOSE AGE IS NOT 20

```
125      -- Display first 3 characters of each first name
126 •    SELECT FirstName,
127      SUBSTRING(FirstName, 1, 3) AS ShortName
128      FROM Student;
```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	FirstName	ShortName
▶	Rahul	Rah
	Priya	Pri
	Amit	Ami
	Neha	Neh
	Rohit	Roh
	Anjali	Anj
	Vikas	Vik
	Sneha	Sne
	Arjun	Arj
	Kavita	Kav
	Saurabh	Sau
	Pooja	Poo

COUNT MALE STUDENTS

```
130      -- Display average age rounded to 2 decimals
131 •    SELECT ROUND(AVG(Age), 2) AS AverageAge
132      FROM Student;
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content

	AverageAge
▶	20.94

COUNT FEMALE STUDENTS

```
134 -- Count students for each age
135 • SELECT Age, COUNT(*) AS TotalStudents
136 FROM Student
137 GROUP BY Age
138 ORDER BY Age;
```

Result Grid |   Filter Rows: | Export:  | Wrap

	Age	TotalStudents
▶	19	5
	20	15
	21	13
	22	12
	23	5

SORT STUDENTS BY ADMISSION DATE

```
140  -- Find students whose first name starts with 'R'
141  • SELECT * FROM Student
142  WHERE FirstName LIKE 'R%';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

	StudentID	FirstName	LastName	Gender	Age	Email	PhoneNumber	Course	AdmissionDate
▶	1	Rahul	Sharma	Male	20	rahul@email.com	9876543210	BCA	2025-06-01
	5	Rohit	Gupta	Male	20	rohit@email.com	9876543214	MCA	2025-06-05
	14	Riya	Saxena	Female	20	riya@email.com	9876543223	B.Com	2025-06-14
	35	Rakesh	Singh	Male	22	rakesh@email.com	9876543244	B.Com	2025-07-05
	42	Ritika	Soni	Female	20	ritika@email.com	9876543251	BCA	2025-07-12
	49	Rajat	Jain	Male	22	rajat@email.com	9876543258	MCA	2025-07-19

CONCLUSION

This project has helped me build a strong foundation in SQL by solving practical database problems. So far, I have completed 60 SQL queries and plan to continue working toward advanced SQL concepts to enhance my data analytics skills.

The background features a white canvas with decorative elements. In the top-left corner, there is a light blue watercolor splash. The top-right corner is adorned with a solid blue abstract shape. The bottom-left corner contains a solid blue abstract shape, and the bottom-right corner features a light blue watercolor splash. Centered on the canvas is the text 'iTHANK you!' in a bold, dark blue font. The word 'i' is lowercase, while 'THANK' is uppercase. 'you!' is in a lowercase, rounded font style.

iTHANK you!